

EXPERIMENT 8 – MINI PROJECT – McDONALD'S FINANCIAL STATEMENTS DATA ANALYTICS

Domain and Problem Statement:

Domain: Financial Analytics & Business Intelligence

Problem Statement:

McDonald's financial performance changes over time due to variations in revenue, earnings, profitability, market valuation, assets, debt, and dividends. This project addresses these challenges by transforming historical financial statement data into structured business intelligence using **Python, MySQL, Power BI, and Tableau** to evaluate financial performance, profitability, valuation, and financial trends.

Dataset Description:

Dataset Name: McDonald's Financial Statements Dataset

Granularity: Each row represents the financial information of McDonald's for a particular year.

Timeframe Covered: 2002–2022

Key Entities & Attributes:

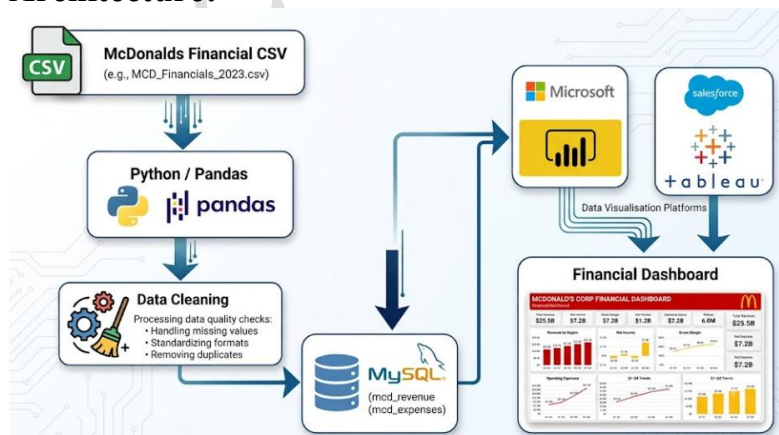
- **Financial Performance:** Revenue, Earnings, EPS, Operating Margin
- **Market Valuation:** Market Cap, P/E Ratio, P/S Ratio, P/B Ratio
- **Financial Position:** Cash on Hand, Net Assets, Total Assets, Total Debt, Total Liabilities
- **Shareholder Information:** Shares Outstanding, Dividend Yield, Dividend

Data Source:

Primary Source: McDonald's historical financial statements dataset provided for the mini project.

Storage Infrastructure: Cleaned financial records can be stored in a local MySQL database.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd
df = pd.read_csv("McDonalds_financial_statements.csv")
mcd = df.copy()
mcd.drop_duplicates(inplace=True)
print("Missing Values:")
print(mcd.isnull().sum())
mcd.columns = (
    mcd.columns
    .str.lower()
    .str.strip()
    .str.replace(" ", "_")
    .str.replace("(", "", regex=False)
    .str.replace(")", "", regex=False)
    .str.replace("$", "", regex=False)
)
print("Original Shape:", df.shape)
print("Cleaned Shape:", mcd.shape)
display(mcd.head())
```

```
===== FIRST FIVE RECORDS =====
```

	Year	Market cap (\$B)	Revenue (\$B)	Earnings (\$B)	P/E ratio	P/S ratio	P/B ratio	Operating Margin (%)	EPS (\$)	Shares Outstanding (\$B)	Cash on Hand (\$B)	Dividend Yield (%)	Dividend (stock split adjusted) (\$)	Net assets (\$B)
0	2022	193.01	23.18	7.82	31.3	8.33	-32.2	33.76	8.42	0.73	2.58	2.15	5.66	-6.01
1	2021	200.31	23.22	9.12	26.5	8.63	-43.5	39.31	10.11	0.74	4.70	1.96	5.25	-4.61
2	2020	159.88	19.20	6.14	33.8	8.32	-20.4	31.97	6.35	0.74	3.44	2.35	5.04	-7.83
3	2019	147.47	21.28	8.01	24.8	6.93	-18.0	38.04	7.98	0.74	0.89	2.39	4.73	-8.22
4	2018	136.21	21.02	7.81	23.2	6.48	-21.8	37.17	7.65	0.76	0.86	2.36	4.19	-6.26

```
===== SHAPE OF DATASET =====
(21, 17)

===== DATA TYPES =====
Year                                int64
Market cap ($B)                    float64
Revenue ($B)                       float64
Earnings ($B)                      float64
P/E ratio                          float64
P/S ratio                           float64
P/B ratio                           float64
Operating Margin (%)                float64
EPS ($)                            float64
Shares Outstanding ($B)             float64
Cash on Hand ($B)                   float64
Dividend Yield (%)                  float64
Dividend (stock split adjusted) ($) float64
Net assets ($B)                     float64
Total assets ($B)                   float64
Total debt ($B)                     float64
Total liabilities ($B)              float64
```

2. Create Derived Financial Columns

Revenue Growth

```
mcd["revenue_growth"] = mcd["revenue_b"].pct_change() * 100
```

Earnings Growth

```
mcd["earnings_growth"] = mcd["earnings_b"].pct_change() * 100
```

Debt to Asset Ratio

```
mcd["debt_to_asset_ratio"] = (
    mcd["total_debt_b"] / mcd["total_assets_b"]
)
```

```
) * 100
```

```
# Liability to Asset Ratio
```

```
mcd["liability_to_asset_ratio"] = (  
    mcd["total_liabilities_b"] / mcd["total_assets_b"]  
) * 100
```

```
# Display result
```

```
display(mcd.head())
```

If your cleaned column names differ slightly after the standardization step, use the names displayed by `print(mcd.columns)`.

	year	market_cap_b	revenue_b	earnings_b	p/e_ratio	p/s_ratio	p/b_ratio	operating_margin_%	eps_	shares_outstanding_b	...	dividend_yield_%	dividend_stock_split_adjusted_	net_assets_b	total_assets_b	total_debt_b	total_l
0	2022	193.01	23.18	7.82	31.3	8.33	-32.2	33.76	8.42	0.73	...	2.15	5.66	-6.01	50.43	48.03	
1	2021	200.31	23.22	9.12	26.5	8.63	-43.5	39.31	10.11	0.74	...	1.96	5.25	-4.61	53.60	48.64	
2	2020	159.88	19.20	6.14	33.8	8.32	-20.4	31.97	6.35	0.74	...	2.35	5.04	-7.83	52.62	48.51	
3	2019	147.47	21.28	8.01	24.8	6.93	-18.0	38.04	7.98	0.74	...	2.39	4.73	-8.22	47.51	46.87	
4	2018	136.21	21.02	7.81	23.2	6.48	-21.8	37.17	7.65	0.76	...	2.36	4.19	-6.26	32.81	31.07	

3. MySQL – Database and Data Loading

from sqlalchemy import create_engine

```
username = "root"
```

```
password = "YOUR_MYSQL_PASSWORD"
```

```
host = "localhost"
```

```
port = 3306
```

```
# Connect to MySQL
```

```
db_engine = create_engine(  
    f'mysql+pymysql://{username}:{password}@{host}:{port}/mcdonalds_financials'  
)
```

```
# Load cleaned data
```

```
mcd.to_sql(  
    "mcdonalds_financials",  
    con=db_engine,  
    if_exists="replace",  
    index=False  
)
```

```
print("Data loaded successfully into MySQL.")
```

Procedure:

Step 1: Data Preparation & Preprocessing (Python / Pandas)

- Load the McDonald's financial statements CSV file into Jupyter Notebook using `pandas.read_csv()`.
- Inspect the dataset for missing and duplicate values.
- Remove duplicate records and handle missing values.
- Standardize column names and verify numerical data types.
- Create derived financial measures such as revenue growth, earnings growth, and debt-to-asset ratio.
- Store the cleaned dataset for further analysis.

	Year	Market cap (\$B)	Revenue (\$B)	Earnings (\$B)	P/E ratio	P/S ratio	\
0	2022	193.01	23.18	7.82	31.3	8.33	
1	2021	200.31	23.22	9.12	26.5	8.63	
2	2020	159.88	19.20	6.14	33.8	8.32	
3	2019	147.47	21.28	8.01	24.8	6.93	
4	2018	136.21	21.02	7.81	23.2	6.48	

	P/B ratio	Operating Margin (%)	EPS (\$)	Shares Outstanding (\$B)	\
0	-32.2	33.76	8.42	0.73	
1	-43.5	39.31	10.11	0.74	
2	-20.4	31.97	6.35	0.74	
3	-18.0	38.04	7.98	0.74	
4	-21.8	37.17	7.65	0.76	

	Cash on Hand (\$B)	Dividend Yield (%)	Dividend (stock split adjusted) (\$)	\
0	2.58	2.15	5.66	
1	4.70	1.96	5.25	
2	3.44	2.35	5.04	
3	0.89	2.39	4.73	
4	0.86	2.36	4.19	

	Net assets (\$B)	Total assets (\$B)	Total debt (\$B)	Total liabilities (\$B)
0	-6.01	50.43	48.03	56.43
1	-4.61	53.60	48.64	58.20
2	-7.83	52.62	48.51	60.45
3	-8.22	47.51	46.87	55.72
4	-6.26	32.81	31.07	39.06

(21, 17)	
Year	0
Market cap (\$B)	0
Revenue (\$B)	0
Earnings (\$B)	0
P/E ratio	0
P/S ratio	0
P/B ratio	0
Operating Margin (%)	0
EPS (\$)	0
Shares Outstanding (\$B)	0
Cash on Hand (\$B)	0
Dividend Yield (%)	0
Dividend (stock split adjusted) (\$)	0

Step 2: SQL Database Connection & Data Ingestion

- Start the MySQL server.
- Create a database named **mcdonalds_financials**.
- Connect Python to MySQL using SQLAlchemy.
- Upload the cleaned McDonald's financial data into the MySQL table.
- Verify the number of records stored in the database.

localhost:3306: McDonalds_financial_statements [1]

McDonalds_financial_statements.income_statement

metric	FY2023	FY2022	Change %	Notes
Total Revenue	25,493.5 M	23,182.6 M	+10%	Inc. Franchise Fees
Cost of Sales	25,493.5 M	23,182.6 M	+10%	Improved Margin
Operating Income	10,950.0 M	9,500.0 M	+15%	Improved Margin
Operating Income	10,950.0 M	9,500.0 M	+15%	Improved Margin
Net Income	7,800.0 M	6,600.0 M	+18%	Strong Quarter
Total Revenue	8,600.0 M	5,200.0 M	+10%	
Operating Income	3,200.0 M	5,400.0 M	+15%	
Operating Income	8,500.0 M	5,000.0 M	+15%	Strong Quarter
Operating Asset	2,000.0 M	1,300.0 M	+10%	
Total Revenue	3,700.0 M	4,500.0 M	+10%	
Operating Income	2,200.0 M	1,900.0 M	+15%	
Net Income	7,800.0 M	6,600.0 M	+18%	Strong Quarter
Net Income	7,800.0 M	6,600.0 M	+18%	Strong Quarter
EPS	0.12	0.15	-0.2%	

The data in the preview has been updated for McDonalds_financial_statements and truncated due to size limits.

Select Related Tables Load Transform Data Cancel

Step 3: Power BI Data Connection & Preparation

1. Open **Power BI Desktop**.
2. Click **Get Data** → **MySQL Database**.
3. Enter:

Server: localhost:3306

Database: mcdonalds_financials

4. Enter the MySQL credentials.
5. Select the **mcdonalds_financials** table.
6. Click **Load** or **Transform Data**.
7. Verify the data types of Year, Revenue, Earnings, Market Cap, EPS, Debt, Assets, and Liabilities.

Step 4: Core Calculations & Measures (DAX)

Create the following Power BI measures:

Total Revenue

Total Revenue = SUM(mcdonalds_financials[revenue_b])

Total Earnings

Total Earnings = SUM(mcdonalds_financials[earnings_b])

Average EPS

Average EPS = AVERAGE(mcdonalds_financials[eps_])

Average Operating Margin

Average Operating Margin =

AVERAGE(mcdonalds_financials[operating_margin_%])

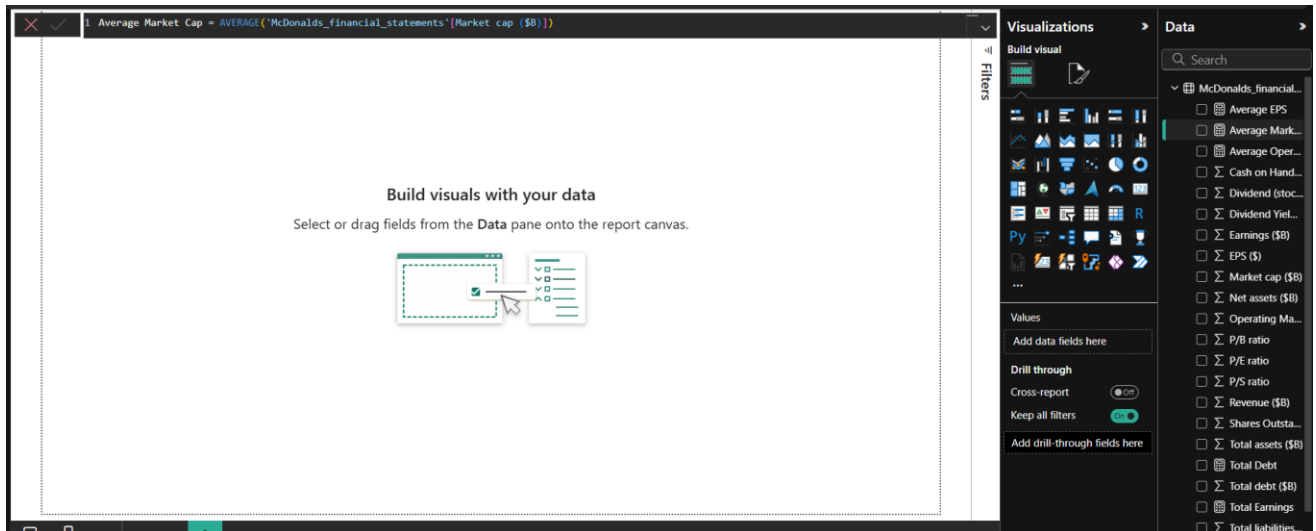
Total Debt

Total Debt = SUM(mcdonalds_financials[total_debt_b])

Average Market Cap

Average Market Cap =

AVERAGE(mcdonalds_financials[market_cap_b])



Step 5: Power BI Financial Dashboard

Create an executive financial dashboard containing:

KPI Cards

- Total Revenue
- Total Earnings
- Average EPS
- Average Operating Margin
- Average Market Cap

Line Chart

Use:

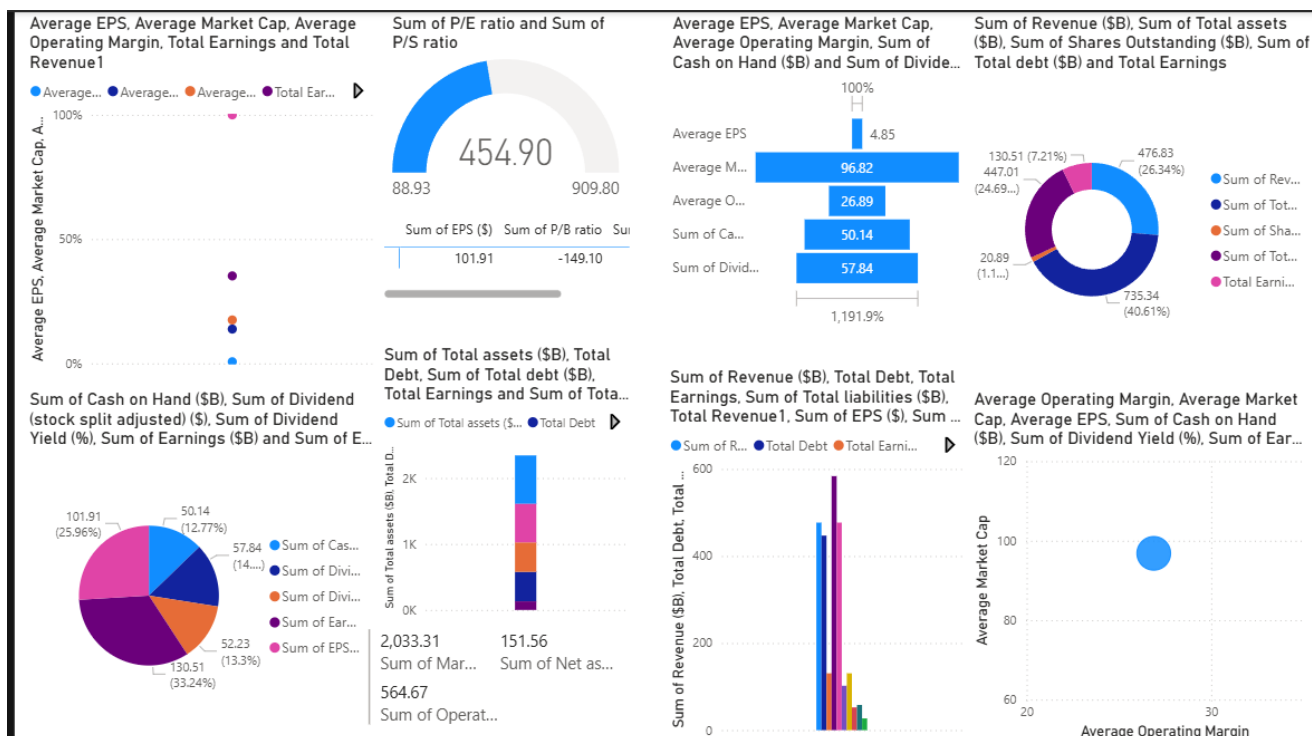
- **Year** → X-axis
- **Revenue** → Y-axis

Bar Chart

- Year → X-axis
- Operating Margin → Y-axis

Step 6: Tableau Data Connection & Preparation

1. Open **Tableau Desktop / Tableau Public**.
2. Select **Text File** or connect to the MySQL database.
3. Select the **McDonald's financial statements** dataset.
4. Verify the financial fields.
5. Create calculated fields for:
 - Total Revenue
 - Total Earnings
 - Average EPS
 - Average Operating Margin
 - Debt-to-Asset Ratio



Step 7: Tableau Financial Dashboard & Interactive Story

Create the following worksheets:

1. KPI Sheet

Display:

- Revenue
- Earnings
- EPS
- Operating Margin

2. Revenue Trend

Create a **Line Chart**

3. Earnings Trend

Create a **Line Chart**:

Earnings → Rows

4. Market Valuation

Create a chart showing:

Year vs Market Cap

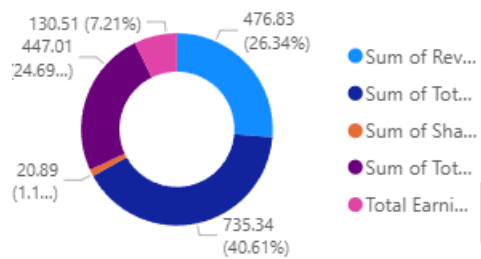
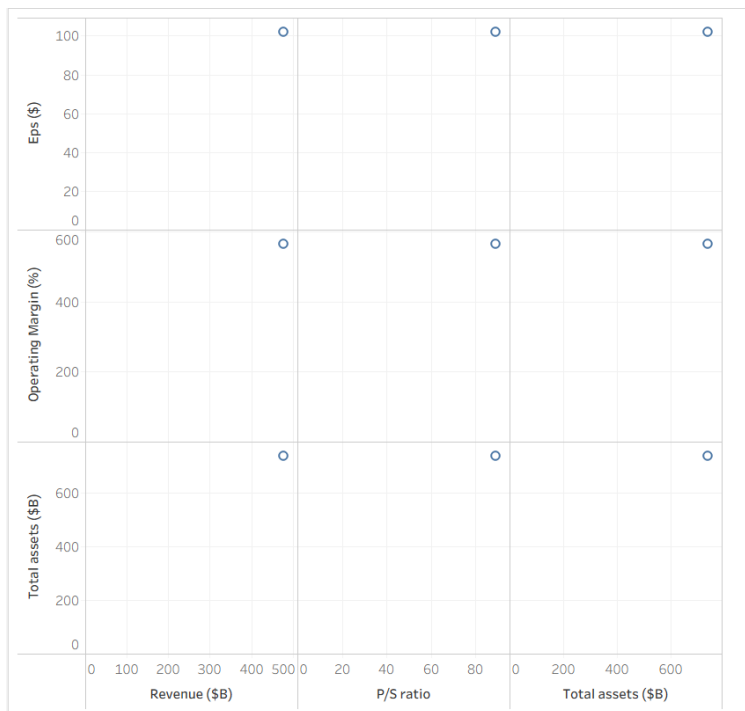
5. Debt and Assets

Create a chart comparing:

Total Assets vs Total Debt

6. Interactive Dashboard

Combine the worksheets into a single dashboard and add **Year** as an interactive filter.



RESULT: